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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE:24AD3107 DATA SCIENCE

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TITLE OF THE PROJECT	Data Quality Scoring Dashboard	
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Project Title: Data Quality Scoring Dashboard

Abstract:

The **Data Quality Scoring Dashboard** is designed to evaluate and monitor the quality of incoming datasets using a standardized quality score. In real-world applications, datasets often contain missing values, inconsistent data formats, duplicate records, and other quality issues that can negatively affect data analysis and decision-making. The proposed system addresses these challenges by automatically analyzing datasets and generating a quality score based on key parameters such as **completeness, consistency, and duplication**.

The system preprocesses the incoming data, identifies quality issues, calculates individual scores for each quality parameter, and combines them into an overall data quality score. A dashboard provides a clear visual representation of the results, allowing users to quickly understand the condition of their datasets and identify areas that require improvement. The system can also display statistics, quality indicators, and detected issues to support effective data cleaning and management.

The dashboard provides detailed insights into the dataset by presenting the number of records, columns, missing values, duplicate records, unique values, and data-type information. Based on these observations, the system generates individual quality scores and an overall percentage score. Different quality levels can be assigned to the dataset, such as **Excellent, Good, Average, or Poor**, making the results easier to interpret. The system can also generate visual charts and graphs to represent quality metrics in a simple and understandable manner.

Furthermore, the proposed system can help organizations establish a consistent approach to data-quality monitoring. Instead of manually checking large datasets, users can upload a dataset and receive an automated assessment within a short period. By identifying data-quality problems before analysis, the system can prevent inaccurate results and improve the performance of data-driven applications. The dashboard can also support continuous monitoring, allowing users to compare the quality of datasets received from different sources or at different time periods.

The proposed solution helps reduce manual data-quality assessment, improves the reliability of datasets, and enables organizations to make better data-driven decisions. Overall, the project demonstrates how **data science, data preprocessing, quality metrics, automated scoring, statistical analysis, and visualization** can be integrated into a practical system for automated data quality assessment. The system can be useful in various domains such as **healthcare, finance, education, e-commerce, business analytics, banking, and customer management**, where accurate, complete, and consistent data is essential for effective decision-making.